CS224

Lab 6

Section 4

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No	Cache Size KB	N Way Cache	Word Size	Block Size	No. of Sets	Tag Size in bits	Index Size (Set No.) in bits	Word Block Offset Size in bits	Byte Offset Size in bits	Block Replacement Policy Needed (Yes/No)
1	64	1	32 bits	4	4096	16	12	2	2	No
2	64	2	32 bits	4	2048	17	11	2	2	Yes
3	64	4	32 bits	8	512	18	9	3	2	Yes
4	64	Full	32 bits	8	1	27	0	3	2	Yes
9	128	1	16 bits	4	16384	15	14	2	1	No
10	128	2	16 bits	4	8192	16	13	2	1	Yes
11	128	4	16 bits	16	1024	17	10	4	1	Yes
12	128	Full	16 bits	16	1	27	0	4	1	Yes

2. Consider the following MIPS code segment. Cache capacity is 8 words, Block size: 2 words, N= 1.

```
        addi $t0, $0, 5
loop:   beq $t0, $0, done
        lw $t1, 0x4($0)
        lw $t2, 0xC($0)
        lw $t3, 0x8($0)
        addi $t0, $t0, -1
        j loop
done:
```

a)

Instruction	Iteration No.				
	1	2	3	4	5
lw $t1, 0x4($0)	Compulsory	Hit	Hit	Hit	Hit
lw $t2, 0xC($0)	Compulsory	Hit	Hit	Hit	Hit
lw $t3, 0x8($0)	Hit	Hit	Hit	Hit	Hit

b)

V	Tag	Data	Data	Set Index
1 bit	27 bits	32 bits	32 bits	3
1 bit	27 bits	32 bits	32 bits	2
1 bit	27 bits	32 bits	32 bits	1
1 bit	27 bits	32 bits	32 bits	0

Cache Capacity = 8 words

Each block = 2 words

So There are 8/2 = 4 blocks.

Since N=1, This is a direct mapped cache which means Set no = block no. Thus there are 4 sets.

log_2^2= 1 bit for block offset.

log_2^4= 2 bit for set index.

log_2^4= 2 bit for the byte offset (32 bit / 8 = 4 byte).

Tag size = 32 - 2 - 2 - 1 = 27.

Cache size = 4 x 27 + 8 x 32 + 4 = 368.

c)

Hardware requirement

One 2:1 Multiplexer is required in order to decide the word in a block

One Equality Comparator is required to decide on if Tags are compatible.

One AND gate is needed to decide HIT.

3. Consider the above MIPS code segment. The cache capacity is 2 words , block size is 1 word. There is only 1 set. The block replacement policy is LRU.

a)

Instruction	Iteration No.				
	1	2	3	4	5
lw $t1, 0x4($0)	Compulsory	Capacity	Capacity	Capacity	Capacity
lw $t2, 0xC($0)	Compulsory	Capacity	Capacity	Capacity	Capacity
lw $t3, 0x8($0)	Capacity	Capacity	Capacity	Capacity	Capacity

b)

Cache representation

V	Tag	Data	V	Tag	Data	Set Index
1 bit	30 bits	32 bits	1 bit	30 bits	32 bits	0

Cache capacity is 2 words

block size = 1 word. so there are 2/1 = 2 blocks.

So we have 1 set with 2 blocks having 1 word in each.

Byte offset = 32 bits / 8 = 4 bytes. log_2^4= 2 bits. so we will have 2 bits for byte offset representation.

Tag size = 32 - 2 = 30 bits.

LRU size = 1 bit.

Thus, the total Cache size with LRU bit = 30 x 2 + 32 x 2 + 1 x 2 + 1 = 127 bits.

c)

Hardware requirement

One 2:1 Multiplexer is required in order to decide on the word in a block

Two Equality Comparators are required to decide on if Tags are compatible.

Two AND gates and one OR gate are needed to decide on HIT.

4)

T_{L1} = 1 clock cycle

T_{L2} = 5 clock cycle

$T_{MainMemory}$ = 55 clock cycle

Miss rate for L1 = 0.2

Miss rate for L1 = 0.05

Amat = 1 + (0.2[5 + 0.05 (55)]) = 1 + 1.55 = 2.55 clock cycles.

He have 2.55 x 10^{12} total clock cycles for instructions

4 GHz = 4 x 10^{9} $cycles/sec.$

so $T_{Exe} = \frac{2.55 \, x \, 10^{12}}{4 \, x \, 10^{9}}$ = 637.5 seconds.